

Dairy intake and semen quality among men attending a fertility clinic

Objective To examine the relationship between dairy food intake and semen parameters **Design** Longitudinal study **Setting** Men attending academic medical center fertility clinic in Boston, MA **Patients** 155 men **Interventions** None **Main Outcome Measures** total sperm count, sperm concentration, progressive motility, and morphology **Results** Low-fat dairy intake was positively related to sperm concentration and progressive motility. On average, men in the highest quartile of intake (1.22–3.54 servings/day) had 33% (95% confidence interval (CI) 1, 55) higher sperm concentration and 9.3 (95%CI 1.4, 17.2) percentage units higher sperm motility than men in the lowest quartile of intake (≤ 0.28 servings/day). These associations were primarily explained by intake of low-fat milk. The corresponding results for low-fat milk were 30% (95%CI 1, 51) higher sperm concentration and 8.7 (95%CI 3.0, 14.4) percentage units higher sperm motility. Cheese intake was associated with lower sperm concentration among ever smokers. In this group, men in the highest tertile of intake (0.82–2.43 servings/day) had 53.2% (95%CI 9.7, 75.7) lower sperm concentration than men in the lowest tertile of cheese intake (< 0.43 servings/day). **Conclusions** Our findings suggest that low-fat dairy intake, particularly low-fat milk, is related to higher sperm concentration and progressive motility, while cheese intake to lower sperm concentration among past or current smokers.